Math 252 [redacted] **Quiz 4** **Name:**

1. Use Part 1 of the Fundamental Theorem of Calculus to find the **derivative** of $F(x) = \int_0^{\sin x} \frac{1}{t^3+1}\,dt$.

$F'(x) = \frac{1}{(\sin x)^3 + 1} \cdot \cos x$

2. Evaluate the following integrals by u-substitution. Make sure to indicate the u-functions.

1). $\int \frac{x^3}{(x^4-2)^2}\,dx$

$u = x^4 - 2$

$du = 4x^3\,dx$

$\left(\frac{1}{4}du = x^3 dx\right)$

$= \frac{1}{4}\int \frac{1}{u^2}\,du$

$= -\frac{1}{4} \cdot \frac{1}{u} + C$

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$= -\frac{1}{4} \cdot \frac{1}{x^4 - 2} + C$

2). $\int \cos^5\theta \sin\theta\,d\theta$

$u = \cos\theta$

$du = -\sin\theta\,d\theta$

$= -\int u^5\,du$

$= -\frac{1}{6}u^6 + C$

$= -\frac{1}{6} \cdot (\cos\theta)^6 + C$